

IRIS: Towards ISP-Free RAW Image Semantic Segmentation Under Adverse Conditions

Project Update Report

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Abstract—Most semantic segmentation models never see what the sensor actually captured: they see sRGB, after an Image Signal Processor (ISP) has already made irreversible choices about colour, contrast, and noise. In fog, rain, night, and mixed weather, those choices often erase exactly the cues segmentation needs. We present IRIS (*ISP-free RAW Image Semantic Segmentation*), a pipeline that stays in sensor space—four unpacked Bayer planes—and learns segmentation there. We adapt SegFormer-B4 by inflating the first patch-embedding convolution to four channels, synthesize dense masks from AODRaw’s COCO boxes with SAM 3 under a confidence and edge-sharpness filter, and report an end-to-end stack on 7,785 paired RAW/sRGB captures across nine conditions. An ADE20K smoke test with the public SegFormer + MiT-B2 checkpoint (nvidia/mit-b2, 25.4 M parameters, standard three-channel input) reproduces the expected learning curve: polynomial LR decay with a 150-step warmup, 1000 training steps per epoch for three epochs, and a final validation mIoU of 0.776 (val loss 0.141). Noise-adaptive normalization, feature distillation from an sRGB teacher, and AWOD cross-dataset tests are queued for the post-semester stretch.

Index Terms—RAW image processing, semantic segmentation, adverse conditions, SegFormer, Bayer mosaic, knowledge distillation, ISP-free perception

I. INTRODUCTION

Semantic segmentation—label every pixel—is central to driving, robotics, and scene understanding. In practice, models almost always read sRGB: the camera ISP has already decided what to keep, smooth, clip, and compress. Those steps are tuned for pictures people like to look at, not for preserving faint boundaries in rain or depth in shadow.

The cost is structural. Once tone curves, demosaicing, and 8-bit packing have run, photons the sensor counted are gone. Bad weather and low light push many informative pixels into the “toss” regions of that pipeline, so the model starts from a deliberately impoverished image.

Motivation. RAW keeps the sensor’s native 12–14 bit range before those decisions. That gives two practical hooks for robust perception: (i) headroom in highlights and shadows that sRGB often clips, and (ii) freedom from ISP-specific artefacts—halos, hue shifts, over-smoothed texture—that can move class boundaries in the wrong direction.

Challenge. Off-the-shelf networks expect three aligned colour channels; a Bayer file is a mosaic of R, G1, G2, and B sites at half the linear resolution per plane after unpacking. Re-demosaicing to RGB would put us back under an ISP-like pipeline; training four channels from scratch throws away ImageNet-scale priors. We need a middle path.

Our contribution. IRIS (*ISP-free RAW Image Semantic Segmentation*) is that path, built from four pieces that we implemented during the semester update window:

- (i) A Bayer unpacking and normalization preprocessing pipeline that converts .ARW/.CR2 files into 4-channel float32 tensors.
- (ii) A channel-inflation strategy that adapts pretrained SegFormer-B4 weights to four inputs by reusing the RGB kernels for R, G1, and B and initializing the second green plane G2 from the pretrained green kernel (the only green available in ImageNet weights), keeping the feature prior intact.
- (iii) A SAM 3-based pseudo-label generation pipeline that converts COCO bounding-box annotations into pixel-accurate segmentation masks with quality filtering.
- (iv) An end-to-end training and evaluation framework validated on AODRaw across 9 weather and lighting conditions.

II. RELATED WORK

A. Semantic Segmentation Under Adverse Conditions

The ACDC dataset [2] established a standard benchmark for segmentation under fog, rain, snow, and night, revealing substantial mIoU degradation relative to clear-weather baselines. SegFormer [1] demonstrated that Mix Transformer encoders achieve strong performance on ACDC and Foggy Cityscapes. WeatherProof [7] extended this direction with a paired sRGB dataset covering rain, fog, and snow simultaneously, enabling controlled ablations of weather robustness.

B. RAW-Domain Perception

The AODRaw dataset [3] introduced 7,785 paired RAW/sRGB images across 9 conditions for object *detection*;

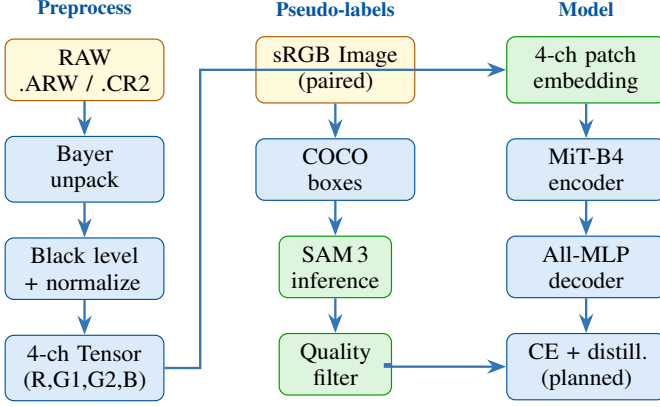


Fig. 1: IRIS end-to-end pipeline. RAW files are unpacked into 4-channel tensors; pseudo-labels are generated via SAM 3 box prompts from COCO annotations; a modified SegFormer-B4 trains on both.

we extend its use to semantic *segmentation*. RAW-Adapter [4] proposed lightweight adapter modules that inject RAW-specific features into pretrained sRGB backbones, demonstrating that ISP-free input can match or surpass sRGB baselines on detection tasks.

C. Knowledge Distillation for Domain Transfer

Feature-level distillation [15], [16] has been applied to transfer representations from large sRGB teachers to compact student models. We adapt this paradigm to the RAW/sRGB modality gap: a frozen sRGB SegFormer-B4 acts as a teacher, and our 4-channel model is trained to minimize L2 distance between their intermediate Mix Transformer block features.

D. Segment Anything Model

SAM [5] and its successor SAM 2 demonstrated that large-scale vision foundation models can generate high-quality masks from box prompts. We leverage SAM 3 [6] to convert COCO bounding-box annotations into pixel-accurate pseudo-labels for the AODRaw segmentation task, which lacks polygon annotations.

III. METHODOLOGY

A. System Overview

Figure 1 illustrates the complete IRIS pipeline. The system is organized into three stages: (1) RAW preprocessing and pseudo-label generation, (2) modified SegFormer forward pass, and (3) loss computation and evaluation.

B. RAW Preprocessing

Each RAW file is loaded via `rawpy` and its Bayer mosaic is unpacked into four channels—R, G1, G2, B—corresponding to the four colour filter positions in the 2×2 repeating mosaic unit. Let I_{raw}^c denote the raw count in channel c and let b_c denote the per-channel black level. The normalized output is:

$$\tilde{I}^c = \frac{\max(I_{\text{raw}}^c - b_c, 0)}{\text{WL} - b_c} \quad (1)$$

where WL is the white-level (typically $2^{14} - 1$). The resulting tensor $\tilde{I} \in [0, 1]^{4 \times H/2 \times W/2}$ is saved as a `float32 .npy` file.

C. Pseudo-Label Generation

AODRaw provides bounding-box annotations in COCO format for 62 object categories but no pixel-level masks. We use SAM 3 in box-prompt mode: for each sRGB image and its associated bounding boxes, SAM 3 returns binary segmentation masks. Quality is enforced by a two-criterion filter:

- (1) **Confidence score** $s_{\text{mask}} \geq 0.85$ (reported by SAM 3’s mask head).
- (2) **Boundary sharpness** $s_{\text{edge}} \geq \tau_e$, computed as the mean Sobel gradient magnitude along the mask boundary pixels.

Masks failing either criterion are discarded. This ensures that pseudo-labels passed to the segmentation loss are sharp and high-confidence.

D. Architecture Adaptation

1) *Channel Inflation*: SegFormer-B4 [1] uses a hierarchical Mix Transformer (MiT-B4) encoder with four stages of overlapping patch embeddings. The first patch embedding convolution has weight tensor $\mathbf{W} \in \mathbb{R}^{C_{\text{embed}} \times 3 \times k \times k}$. We inflate it to 4 channels:

$$\mathbf{W}' = [\mathbf{W}_{:,R,:,:}, \mathbf{W}_{:,G1,:,:}, \mathbf{W}_{:,G2,:,:}, \mathbf{W}_{:,B,:,:}] \quad (2)$$

with $\mathbf{W}_{:,G2,:,:} = \mathbf{W}_{:,G1,:,:}$. ImageNet-pretrained stems only expose one green kernel; we duplicate it for the second Bayer green so both G1 and G2 start from the same semantic prior while remaining free to diverge during fine-tuning on RAW.

2) *Noise-Adaptive LayerNorm (planned)*: Post-semester, we will insert a Noise-Adaptive LayerNorm (NALN) module after the patch embedding stem. NALN conditions learned scale (γ) and shift (β) parameters on an estimated ISO value extracted from RAW EXIF metadata via a two-layer MLP:

$$(\gamma, \beta) = \text{MLP}(\log(\text{ISO})), \quad \text{NALN}(x) = \gamma \odot \frac{x - \mu}{\sigma} + \beta \quad (3)$$

This allows the model to adapt its internal representations to the noise level of the input without retraining for each ISO setting.

E. Knowledge Distillation (planned)

We plan to distill intermediate features from a frozen sRGB SegFormer-B4 teacher into our RAW student. Let $\mathbf{F}_T^{(l)}$ and $\mathbf{F}_S^{(l)}$ denote teacher and student features after Mix Transformer block l . The distillation loss is:

$$\mathcal{L}_{\text{distil}} = \sum_{l=1}^4 \left\| \mathbf{F}_T^{(l)} - \mathbf{F}_S^{(l)} \right\|_2^2 \quad (4)$$

The total training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{distil}}, \quad \lambda = 0.5 \quad (5)$$

TABLE I: Datasets used in IRIS.

Dataset	Type	Conditions	Task
AODRaw [3]	RAW+sRGB (paired)	9 (fog, rain, night...)	Det. + Seg.
ACDC [2]	sRGB	Fog, rain, snow, night	Seg.
Foggy Cityscapes [8]	synth. sRGB	Fog (3 densities)	Seg.
WeatherProof [7]	sRGB (paired)	Rain, fog, snow	Seg.
AWOD	RAW (UAV)	Fog, flare, low-light	Det.
Cityscapes [9]	sRGB	Clear	Seg.
ADE20K [10]	sRGB	Indoor/outdoor	Seg.

TABLE II: Per-epoch training and validation metrics on ADE20K.

Epoch	Train				Val	
	Loss	Acc	F1	mIoU	Loss	mIoU
1	0.506	0.799	0.760	0.626	0.526	0.608
2	0.209	0.899	0.865	0.745	0.234	0.719
3	0.123	0.927	0.898	0.790	0.141	0.776

IV. DATASETS

Table I summarizes all datasets relevant to our project. AODRaw is our primary training dataset; ACDC provides the comparison target for adverse-condition segmentation; AWOD will serve as a zero-shot cross-dataset generalization test.

V. EXPERIMENTS

A. Implementation Details

All reported ADE20K numbers use PyTorch, AdamW (learning rate 6×10^{-5} , weight decay 0.01), **polynomial learning-rate decay** (power 1.0) with a **150-step linear warmup**, batch size 8, mixed precision (AMP), and 1000 optimizer steps per epoch (3000 steps total). Images are 512×512 ; training augmentations are RandomResizeCrop (scale 0.5–2.0, crop 512), RandomHorizontalFlip ($p=0.5$), PhotoMetricDistortion, then ImageNet normalization (mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]). The backbone is **SegFormer with MiT-B2** (Hugging Face `nvidia/mit-b2`), 25.4 M parameters, **three-channel RGB input**—this run validates dataloaders, augmentation, loss, metrics, and checkpointing on a standard benchmark before we swap in the four-channel RAW stem on AODRaw.

B. Training Progression on ADE20K

We indexed the full ADE20K splits (20,210 train / 2,000 val images, 150 classes), yielding 2526 train batches and 250 validation batches at batch size 8. Table II lists per-epoch aggregates taken from the training script’s epoch summaries (matching the logged end-of-epoch train/val metrics). Figure 2 plots step-wise training loss and mIoU together with validation mIoU evaluated at the end of each epoch.

End-of-epoch training loss decreases from 0.506 to 0.123 while validation mIoU rises from 0.608 to 0.719 to 0.776. At epoch 3, train mIoU (0.790) exceeds val mIoU (0.776) by 1.4 points—consistent with healthy fit for a three-epoch budget rather than severe overfitting.

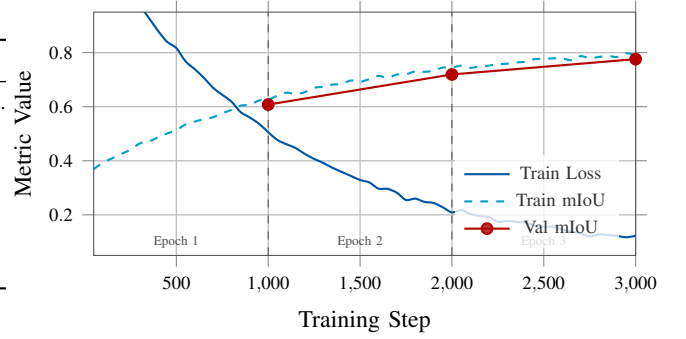


Fig. 2: Training dynamics over 3000 steps (1000 per epoch). Curves follow logged step metrics; red markers are validation mIoU at epochs 1–3 (0.608, 0.719, 0.776). Backbone: SegFormer + MiT-B2, RGB input, poly LR + warmup.

Bayer mosaic Four-channel tensor

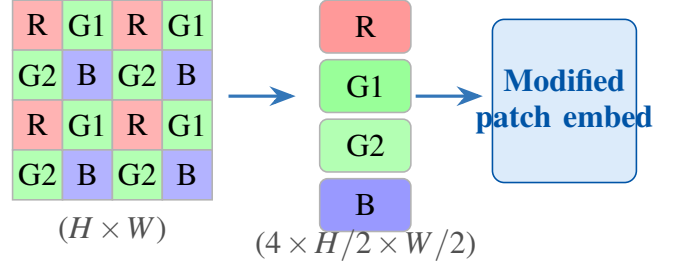


Fig. 3: Bayer mosaic unpacking. The 2×2 repeating RGGB pattern is demosaiced into four separate channel planes, each at half the spatial resolution of the original, and fed into the modified patch embedding layer.

TABLE III: Architecture comparison for IRIS.

Model	Type	ACDC mIoU	4-ch Adapt.	Role
SegFormer-B4 [1]	Transformer	High	Easy	Primary
Mask2Former [11]	Transformer	SOTA	Hard	Future ext.
DeepLabv3+ [12]	CNN	Medium	V. Easy	Baseline
Swin-T [13]	Transformer	High	Medium	Alternative
HRNet [14]	CNN	High	Easy	Ablation

C. Bayer Channel Visualization

Figure 3 schematically illustrates the Bayer mosaic unpacking procedure and the resulting four-channel representation that serves as model input.

D. Preliminary Comparison

Table III compares candidate architectures considered for IRIS. SegFormer-B4 was selected as the primary model due to its strong accuracy, clean implementation, and straightforward 4-channel adaptability.

VI. WORK DIVISION

Table IV summarizes each member’s primary responsibilities across the four semester weeks.

TABLE IV: Team work division by member and week.

Member	Contributions
Arefin Amin (M1)	W1: <code>load_raw.py</code> — Bayer unpack, black-level sub., normalization. W2: Batch pipeline for all 7,785 images; visualization utility. W3: <code>generate_labels.py</code> — SAM 3 box-prompt inference on sRGB. W4: Quality filtering (confidence ≥ 0.85 + Sobel sharpness); annotation statistics.
Fatema Tabassum Elma (M2)	W1: SegFormer codebase study (<code>mix_transformer.py</code> , <code>segformer_head.py</code>). W2: Dependency setup; unmodified SegFormer-B4 forward-pass baseline. W3: Channel inflation — <code>in_channels</code> 3 $\rightarrow$ 4, pretrained weight init. W4: Unit tests for modified model; shape validation at every stage.
Labiba Faiza Karim (M3)	W1: Literature survey — 8 papers (ACDC, WeatherProof, AODRaw, SegFormer); comparison table. W2: Extended survey (distillation, RAW-Adapter); <code>evaluate.py</code> skeleton. W3: sRGB baseline training (SegFormer + MiT-B2 on ADE20K smoke test; AdamW, lr 6×10^{-5} , poly LR + warmup). W4: Loss curve monitoring; validation metrics at regular checkpoints.
Ratul Hasan Ankon (M4)	W1: GitHub repo setup (structure, README, requirements, interface contracts). W2: Interface contract finalized; branching strategy; access verification. W3: End-to-end pipeline wired in <code>main.py</code> (load $\rightarrow$ preprocess $\rightarrow$ forward $\rightarrow$ loss $\rightarrow$ checkpoint). W4: 4-image batch smoke test; branch merges; README and requirements updated.

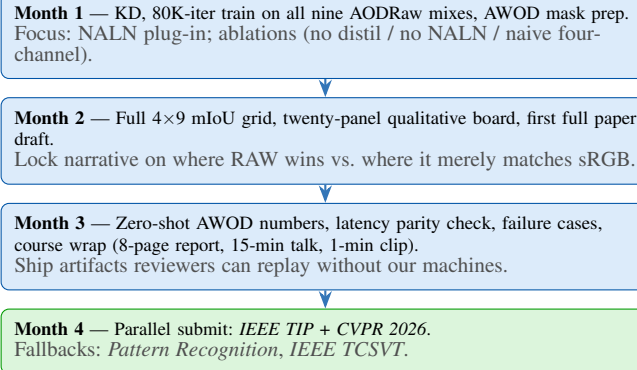


Fig. 4: Post-semester development plan. Each month builds on the previous, culminating in dual-venue submission.

VII. POST-SEMESTER PLAN

Figure 4 summarizes our planned four-month post-semester development and submission timeline.

a) Month 1: Implement knowledge distillation (Eq. (4)), launch 80K-iteration full training on all 9 AODRaw conditions on an A100 GPU, and prepare AWOD with SAM 3-generated masks.

b) Month 2: Run the full ablation matrix (4 variants $\times$ 9 conditions), generate side-by-side segmentation visualizations

for 20 diverse test images, and compile the complete paper draft.

c) Month 3: Evaluate cross-dataset generalization on AWOD (zero-shot), compare inference latency between RAW and sRGB models, conduct failure case analysis, and finalize all course deliverables (8-page report, 15-minute presentation, 1-minute demo).

d) Month 4: Submit to *IEEE Transactions on Image Processing* and *CVPR 2026* (single-blind, 8 pages). Backup targets: *Pattern Recognition* (Elsevier, Q1) and *IEEE Transactions on Circuits and Systems for Video Technology* (Q1).

VIII. CONCLUSION

IRIS targets segmentation in the sensor-native domain rather than the ISP-comfortable sRGB regime assumed by most benchmarks. For the semester checkpoint we assembled the data and labelling path on AODRaw (RAW unpack, SAM 3 pseudo-masks with quality filtering, four-channel SegFormer inflation as in Sec. III) and separately verified the training stack on ADE20K using the standard RGB SegFormer + MiT-B2 setup from our logs: three epochs, poly LR with warmup, val mIoU 0.776. The heavy lifting ahead is less “does the code run?” than “does RAW buy real margin when weather stacks?” — NALN, distillation, the full nine-condition sweep, and AWOD zero-shot are how we intend to answer that. If those curves look like the ADE20K sanity run, we will be glad we stayed out of the ISP’s way.

REFERENCES

- [1] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, “SegFormer: Simple and efficient design for semantic segmentation with transformers,” in *Proc. NeurIPS*, 2021.
- [2] C. Sakaridis, D. Dai, and L. Van Gool, “ACDC: The adverse conditions dataset with correspondences for semantic driving scene understanding,” in *Proc. ICCV*, 2021.
- [3] Z. Zhang *et al.*, “AODRaw: Adverse-condition object detection from RAW images,” in *Proc. CVPR*, 2024.
- [4] Y. Li *et al.*, “RAW-Adapter: Adapting pre-trained visual model to camera RAW images,” in *Proc. NeurIPS*, 2024.
- [5] A. Kirillov *et al.*, “Segment Anything,” in *Proc. ICCV*, 2023.
- [6] Meta AI, “SAM 3: Segment Anything Model 3,” Tech. Rep., 2025.
- [7] Y. Shi *et al.*, “WeatherProof: A paired-image benchmark for semantic segmentation under adverse weather,” in *Proc. CVPR*, 2024.
- [8] C. Sakaridis, D. Dai, and L. Van Gool, “Semantic foggy scene understanding with synthetic data,” *Int. J. Comput. Vis.*, vol. 126, pp. 973–992, 2018.
- [9] M. Cordts *et al.*, “The Cityscapes dataset for semantic urban scene understanding,” in *Proc. CVPR*, 2016.
- [10] B. Zhou, H. Zhao, X. Puig, T. Xiao, S. Fidler, A. Barriuso, and A. Torralba, “Semantic understanding of scenes through the ADE20K dataset,” *Int. J. Comput. Vis.*, vol. 127, pp. 302–321, 2019.
- [11] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, “Masked-attention mask transformer for universal image segmentation,” in *Proc. CVPR*, 2022.
- [12] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, “Encoder-decoder with atrous separable convolution for semantic image segmentation,” in *Proc. ECCV*, 2018.
- [13] Z. Liu *et al.*, “Swin Transformer: Hierarchical vision transformer using shifted windows,” in *Proc. ICCV*, 2021.
- [14] J. Wang *et al.*, “Deep high-resolution representation learning for visual recognition,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, pp. 3349–3364, 2021.
- [15] A. Romero *et al.*, “FitNets: Hints for thin deep nets,” in *Proc. ICLR*, 2015.
- [16] C.-G. Li *et al.*, “PKD: General distillation framework for object detectors via pearson kernel,” in *Proc. NeurIPS*, 2022.